Committee Reports to the Mechanical Division of the American Railway Association

Lubrication of Cars and Locomotives

While the hydrostatic lubricator with single or double piping is in most general use, the force feed or mechanical lubricator is being used extensively. It has been found that the mechanical lubricator provides more efficient feeding of oil to the valves and cylinders, thereby prolonging the life of valve rings and bushings, and piston rings and bushings. To obtain economy in the amounts of oil consumed, as compared with hydrostatic lubricators, it is necessary to provide a variable feed through a proper connection of the ratchet lever connecting rod, so that as the cut-off is lengthened, the amount of oil fed per revolution of the driving wheels is increased.

Side and Main Rod Bearings

Grease is in general use for side and main rod bearing lubrication, with oil for valve gear bushings and pins. A number of lines are experimenting with a pressure system for side and main rod bearings and motion work lubrication. Reports indicate this system has given satisfactory results, and is an improvement over the old style rod cup for crank pin lubrication, and oil for motion work.

Guides and Piston Rods

Light oil, fed by gravity, is used most for guide lubrication, and properly designed oil cups and oil of proper consistency are important factors in successful lubrication. Some roads are experimenting with connecting feeds from mechanical lubricators to guides, thus lubricating them with valve oil. Reports indicate this method as satisfactory and more efficient than the oil cup method. Some roads, claiming the pistons secure sufficient lubrication from the cylinders, have eliminated piston swabs. An objection to the piston swab is that it might deceive the locomotive man into believing that the cylinder is being lubricated properly when it is not.

Locomotive Trucks

Two general practices are followed in packing locomotive truck cellars, viz., placing the packing in rolls with the ends turned under, or applying in one mass with the ends turned under. These methods appear equally satisfactory. A new departure is being tried with a bearing similar to that of floating bushings used in main rods, the halves of the bearing being bolted together, with the flanges forming the hub liner, and the bearing floating in the box. A grease cup is installed on the truck box, and journal compound is forced to the bearing in a manner similar to that used on ordinary rod cups. Where truck frames with outside bearings and boxes can be applied, they are proving more satisfactory than the inside bearings. Experiments show that where oil holes are drilled straight through the top of locomotive truck brasses, the tendency is for the oil to be forced back out through the top of the brass instead of following around the bearing. Proper results are secured by eliminating the holes and providing oil channels leading around the side of the brass, so that oil poured on top will flow into the cellars and then be fed from the packing.

Driving Journals

Grease is used most generally. Most roads use 3/16 in. holes in the perforated plates on road locomotives, and ⅜ in. holes on switching locomotives, but, with the dehydrated grease being used, some roads find that, by reducing the size of the holes from 3/16 to ⅛ in., and allowing a maximum of 50 lb. follower spring tension when the spring is fully compressed, grease consumption is reduced materially. Some roads are using improved designs of driving box cellars, with reduced contact surface of grease on journal, to reduce the grease consumption.

The method for many years to provide shoe and wedge lubrication has been to cut grooves in driving box or wedge, and drill holes from oil pockets in top of box to shoe and wedge faces, the oil pockets being packed with saturated waste. To provide more efficient lubrication, some roads have removed the saturated waste, welded a plate over oil pockets, applied to pressure fitting, and use a soft grease for lubrication. Reports from them indicate this method has proved satisfactory. In some instances grooves have also been drilled from these pockets to hub plates.

To prevent hot driving boxes, resulting from driving brasses gripping journals due to expansion, driving brasses should be bored sufficiently larger than journals to allow for expansion. The bottom edge of driving brass should be chamfered to assist the feeding of the grease.

Some roads have found that economy results by turning all driving journals when a locomotive receives general repairs. The committee believes this to be good practice.

Hubs and Hub Liners

The committee believes that the practice of lubricating these surfaces by pouring oil on them, at terminals or on the road, is not fully efficient under present operating conditions. Experiments indicate that welding a plate over the oil pockets on driving boxes, connecting to hub plates with an oil hole, applying a pressure cup in the top of the plate, and using a soft grease, provides more efficient lubrication.

One road reports the use of heavy graphite grease on locomotive trucks and trailers, together with floating hub liners, to have materially reduced wear. A hole is drilled through the locomotive truck wheel hub, and the grease is forced through this to the hub plates with a threaded plug similar to that used on rod cups. On the trailer boxes that were originally cored with oil channels leading